



Customer & Revenue Intelligence Platform

End-to-End Data Engineering, Analytics, Machine Learning & GenAI



Student

MANOJ-G1 DA



Guide

NITYASHREE MAM

Business Problem & Objectives

The Challenge

- E-commerce data is siloed and difficult to analyze
- Customer churn goes undetected until too late
- Management lacks automated, actionable insights

Our Solution

Automated Pipeline

End-to-end incremental data ingestion with zero manual intervention

Medallion Architecture

Bronze–Silver–Gold layers for progressive data refinement

ML & AI Insights

Churn prediction model
+ GenAI-generated executive summaries

BI Dashboards

Interactive Power BI & Streamlit apps for decision-making

Dataset & Technology Stack

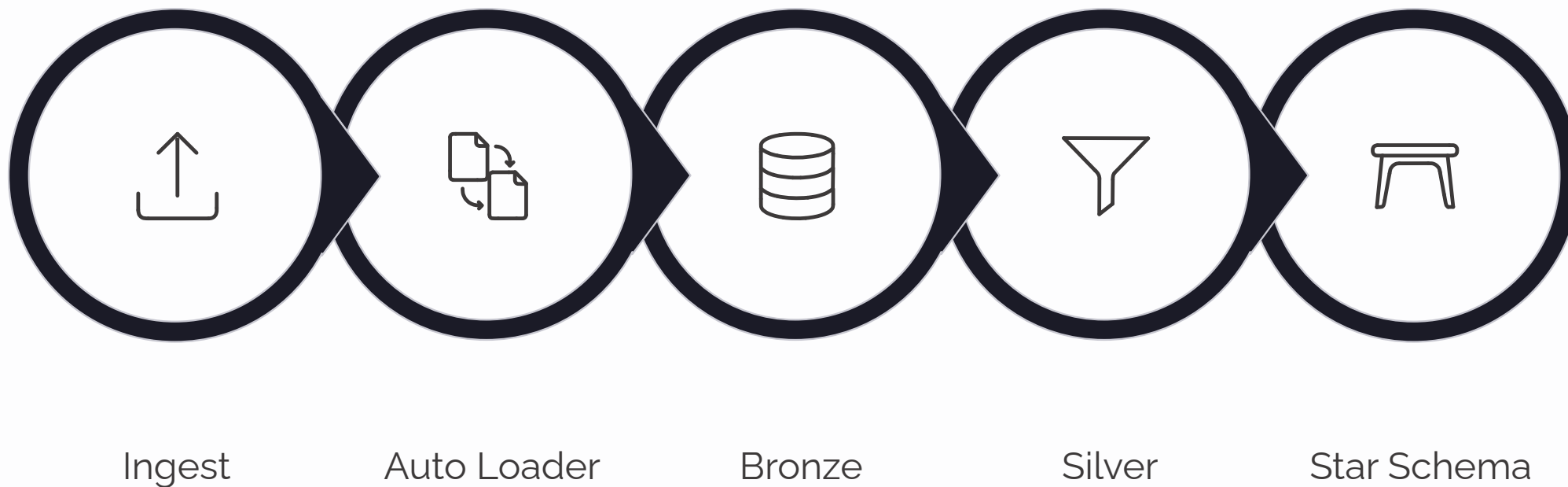
AdventureWorks Dataset

Customers	Products
Sales	Returns
Calendar	Territories

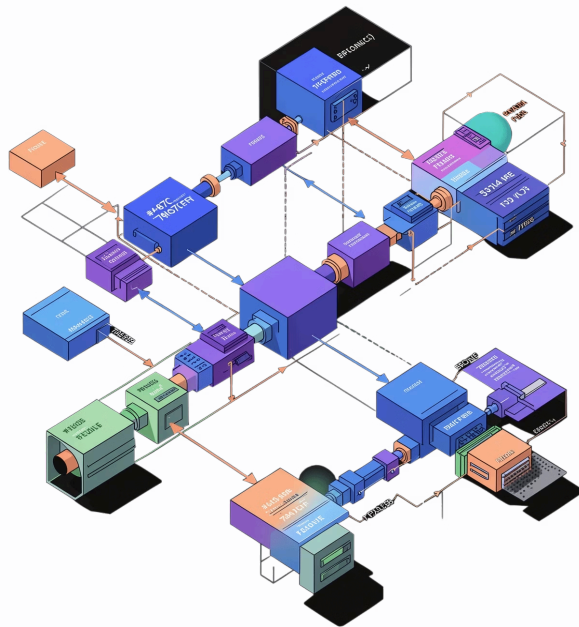
Technology Stack

- Data Engineering
Databricks · PySpark · Delta Lake
- Analytics
SQL · Power BI
- Machine Learning
Spark ML · Random Forest · MLflow
- Applications
Streamlit · Claude AI
- Automation
Databricks Workflows

End-to-End Architecture



*Core Platform: **Databricks + Delta Lake** — orchestrating every stage from raw ingestion to AI-powered insights*



Bronze & Silver Layers

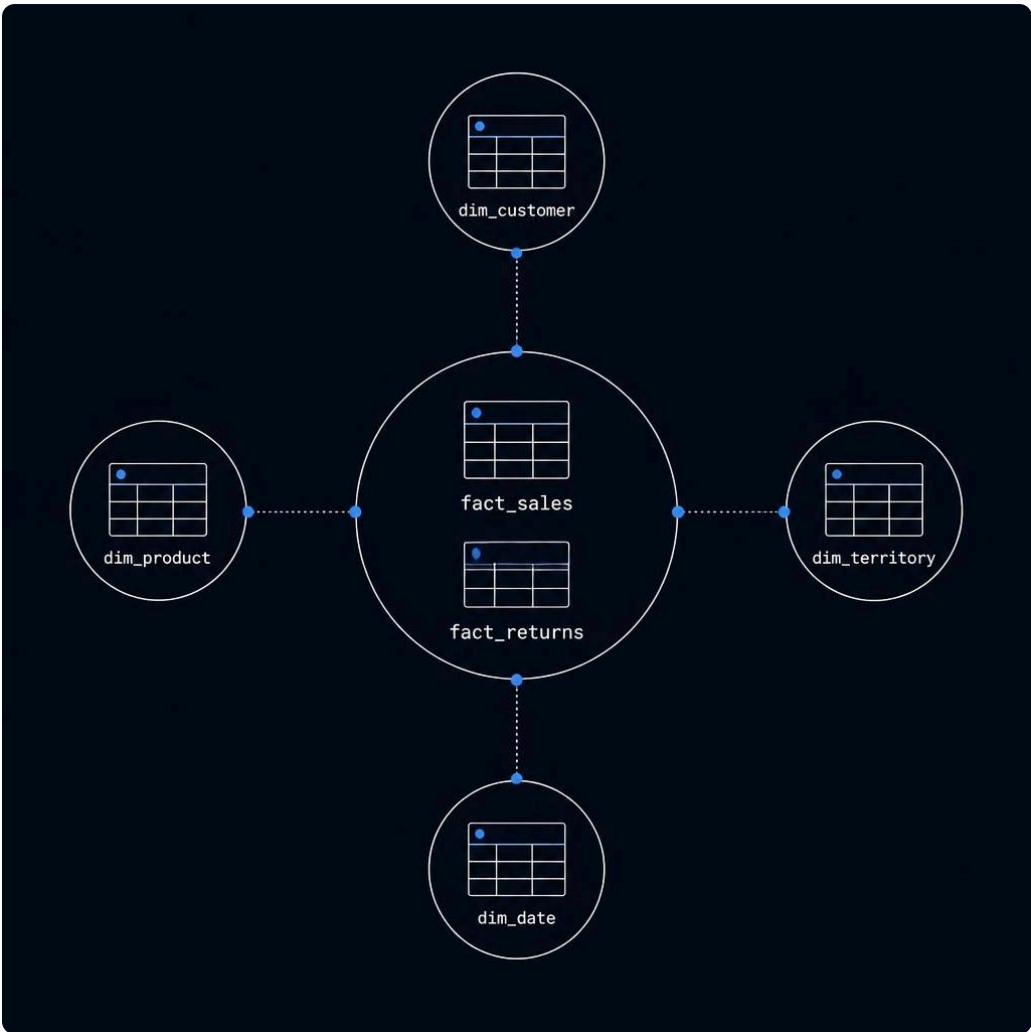
Bronze Layer — Raw Ingestion

- Auto Loader detects & ingests new CSV files automatically
- Incremental checkpoints prevent reprocessing
- Delta tables with MERGE/UPSERT handle updates seamlessly

Silver Layer — Data Quality

- Null/key validation & duplicate removal
- Type conversion & text standardization
- Business rule validation for clean, trusted data

Star Schema & Gold Layer



Star Schema Design

- **Dimensions:** dim_customer, dim_product, dim_territory, dim_date
- **Facts:** fact_sales, fact_returns

Gold Layer — Business-Ready Analytics

gold_kpi

Top-level business metrics

gold_monthly_revenue

Trend & seasonality analysis

gold_customer_summary

Lifetime value & behaviour

gold_product_summary

Category & SKU performance

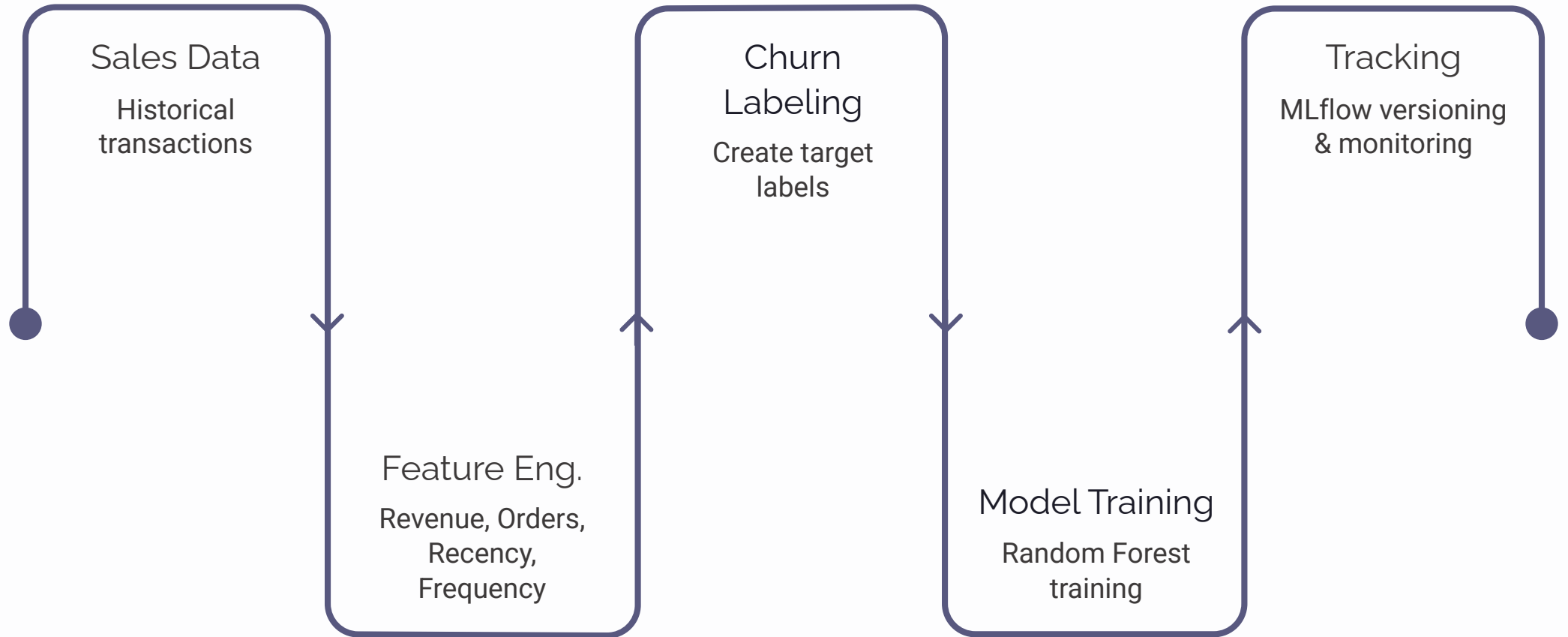
gold_territory_summary

Regional revenue breakdown

gold_return_summary

Return rate & impact

Machine Learning & Churn Prediction

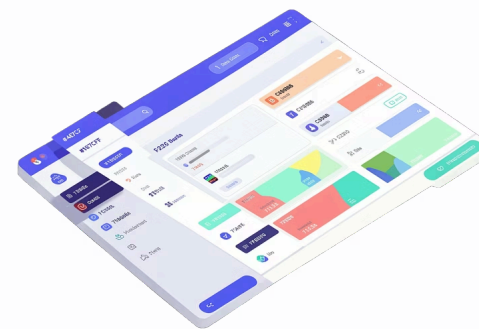


Power BI & Streamlit Applications



Power BI Dashboard

- KPI cards: Revenue, Profit, Orders, Customers
- Revenue trends, category & territory performance
- Churn visualization with interactive filters



Streamlit Application

- Executive Overview & Customer Intelligence
- Product, Territory & Returns Analysis
- AI Business Insights powered by Claude

Both applications draw real-time data directly from **Databricks Gold Tables** — ensuring a single source of truth across all reporting layers.

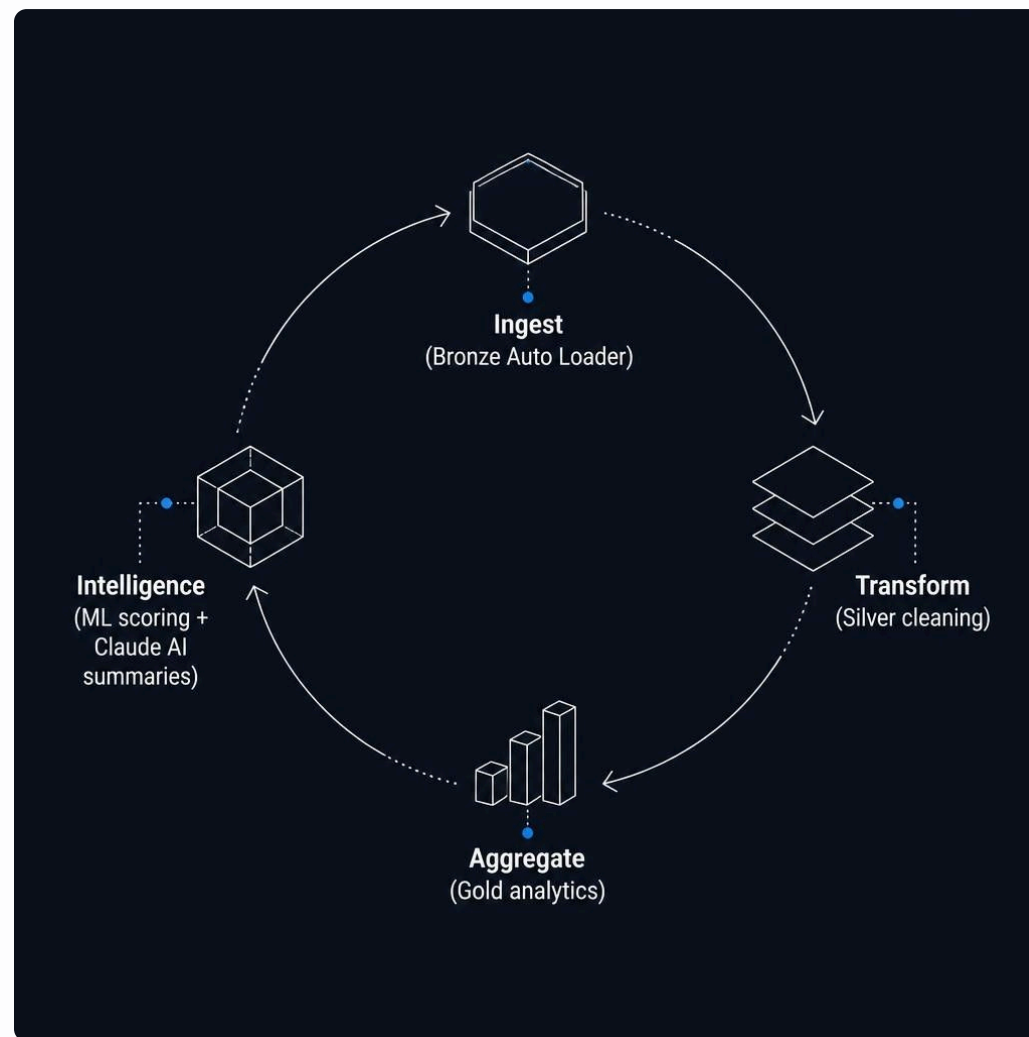
GenAI & Automation

Generative AI — Claude

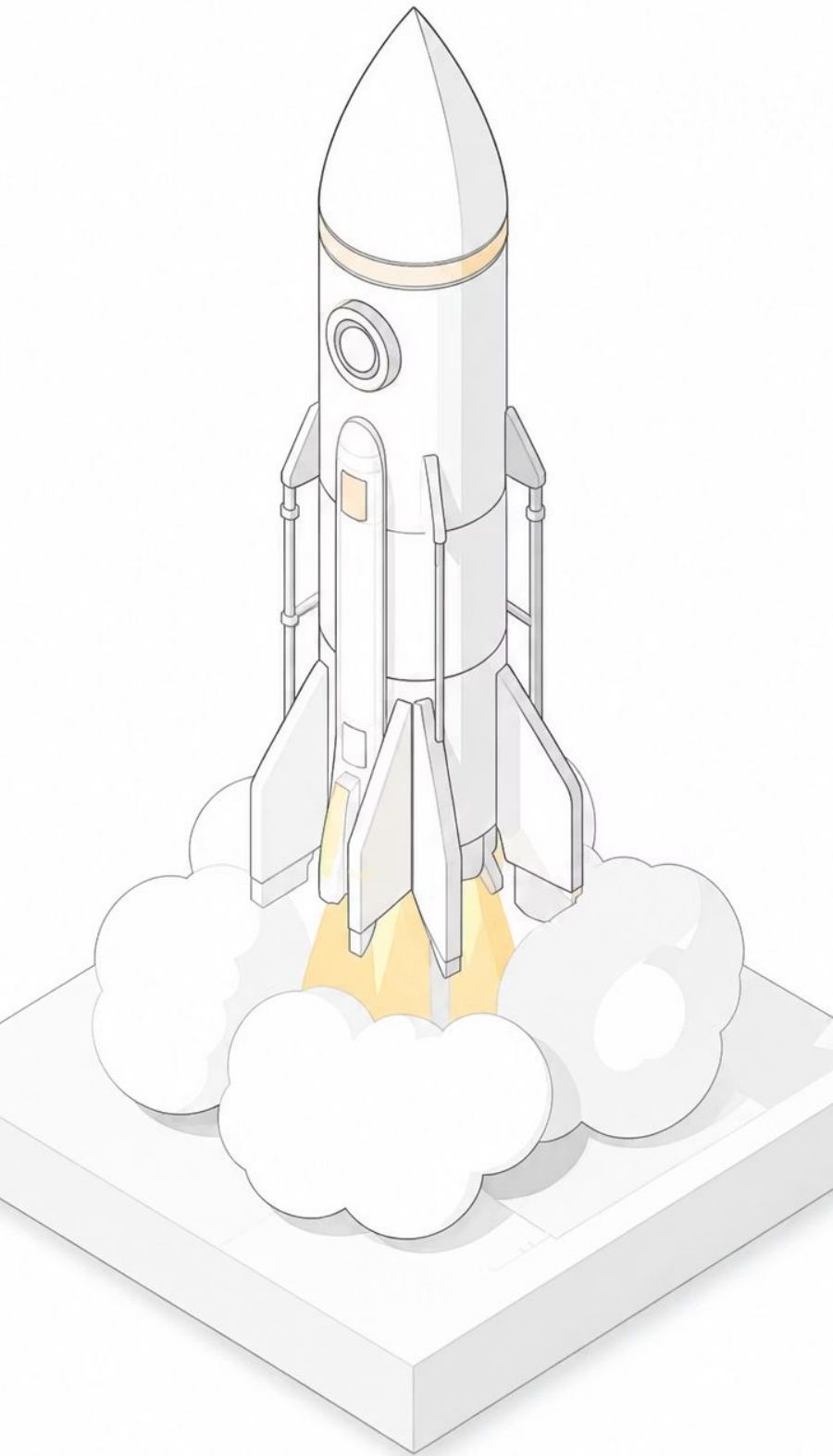
- Analyzes Gold metrics & generates executive summaries
- Revenue, customer, and churn narrative insights
- Actionable business recommendations on demand

Databricks Workflows — Orchestration

- Orchestrates full pipeline: Bronze → Silver → Gold → ML
- Scheduled incremental loads & automated model retraining
- Automated alerts & notifications on failure or anomaly



Conclusion & Future Scope



What I Delivered

- 1 Fully Automated Platform**
End-to-end analytics pipeline with zero manual intervention
- 2 Business-Ready Data Model**
Star Schema with curated Gold layer for all reporting needs
- 3 Intelligence at Scale**
Revenue & customer insights powered by ML & AI
- 4 Predictive Churn Identification**
AI-assisted early warning for customer retention

Future Enhancements

- Real-Time Streaming**
Replace batch ingestion with live event streaming
- Advanced Forecasting**
Anomaly detection & demand forecasting models
- Cloud Deployment**
Production-grade deployment with automated alert pipelines